

Platform Integral Governs Stable Window Formation in Hybrid Modes

Author

Youfu Wang

Independent Researcher

ORCID: 0009-0005-5998-6024

Abstract

Conventional understanding attributes stable window formation in hybrid mode systems to pointwise extrema, such as minimum residue or maximum group index. However, such descriptions fail to explain the observed variation in window width.

Here, using Hopfield coefficient data from polariton systems, we demonstrate that stable windows are governed not by extrema, but by the integrated weight of low-residue plateaus. We introduce a platform integral:

$$P^* = \int \exp(-Z_R / \langle Z_R \rangle) \cdot \exp(-|\partial_{\theta} Z_R| / \langle |\partial_{\theta} Z_R| \rangle) d\theta$$

Across Fig5(d/e/f) datasets, we find:

$$P^*_d < P^*_e < P^*_f$$

which matches the ordering of stable window widths. This establishes that the minimum residue determines the mode center, while the plateau integral governs the window width.

We further outline a cross-system mapping to slow-light photonic crystal systems, suggesting a general platform-based mechanism for window formation.

1. Introduction

Stable transmission or propagation windows appear in a wide range of hybrid mode systems, including polaritonic and photonic crystal structures. Traditional interpretations associate such windows with extrema in system parameters, such as minimum loss or maximum group index.

However, these interpretations are inherently local and fail to explain variations in window width under similar extrema conditions.

This motivates a structural re-examination of how stable windows emerge.

2. Platform Integral Definition

We consider the residue landscape $Z_R(\theta)$ derived from

Hopfield coefficients. Instead of focusing on local minima, we define a global measure of low-residue, low-gradient regions:

$$P^* = \int e^{\{-Z_R/\langle Z_R \rangle\}} \cdot e^{\{-|\partial_{\theta} Z_R|/\langle |\partial_{\theta} Z_R| \rangle\}} d\theta$$

The first term weights low-residue regions, while the second term favors flatness.

3. Data Validation

Using Hopfield coefficient datasets for three configurations (Fig5 d/e/f), we compute:

Case P^*

d 0.7037

e 0.8843

f 0.9318

We observe:

$$P^*_d < P^*_e < P^*_f$$

This ordering matches the observed increase in stable window width.

4. Core Result

We therefore conclude:

The minimum residue determines the mode center, while the platform integral determines the stable window width.

This replaces the traditional extremum-based description with a global structural mechanism.

5. Cross-System Mapping

The same structure can be mapped to slow-light photonic crystal systems:

Polariton Photonic Crystal

$Z_R(\theta) v_g(\lambda)$

plateau flat band

quasi-flat window slow-light window

We propose:

$$P^*_{\text{slow}} = \int e^{\{-v_g \angle v_g\}} e^{\{-|\partial_{\lambda} v_g| \angle |\partial_{\lambda} v_g|\}} d\lambda$$

suggesting a platform-based mechanism across systems.

6. Conclusion

Stable window formation is governed by the integrated structure of low-residue plateaus rather than local extrema.

This provides a new structural perspective on hybrid mode systems and suggests broader applicability across physical platforms.

Keywords

Hybrid modes, polariton, slow light, platform integral, residue landscape, flatband, stability window
